AI Storage Advisor

Overview

AI Storage Advisor — кросплатформна AI-система аналізу дискового простору для Windows та Linux.

Мета проєкту — створити інтелектуального помічника, який не просто показує зайняте місце на диску, а аналізує файлову систему, знаходить зайві дані, оцінює ризики видалення та надає зрозумілі рекомендації за допомогою локальних або хмарних мовних моделей.

Основна відмінність від CCleaner, WizTree та TreeSize — використання AI для пояснення рекомендацій та SRE-аналітики.

Goals

Main Goals

- Аналіз дискового простору
- Пошук зайвих файлів
- Аналіз логів
- Аналіз кешів
- Пошук дублікатів
- Прогнозування заповнення дисків
- AI пояснення рекомендацій
- Аналіз Linux та Windows серверів
- SRE Storage Analytics

Technology Stack

Language

Python 3.12+

GUI

PySide6 (Qt6)

Database

SQLite

Security

cryptography argon2-cffi

Remote Access

paramiko pywinrm

File Processing

pathlib psutil hashlib watchdog

Packaging

PyInstaller AppImage

AI Providers

Local AI

Ollama

Supported Models:

- Qwen
- Gemma
- Llama
- Mistral
- DeepSeek

LM Studio

OpenAI Compatible API

Cloud AI

OpenAI

Anthropic

Gemini

DeepSeek

Custom AI Provider

Support any OpenAI-compatible endpoint.

Settings:

- Base URL
- API Key
- Model
- Temperature
- Context Window

Examples:

- OpenRouter
- Groq
- Together AI
- Fireworks AI
- Azure OpenAI
- Local vLLM
- Corporate AI Gateway

AI First Architecture

All AI providers must work through a single AI abstraction layer.

Business logic must not depend on a specific provider.

Switching providers must not require code changes.

Security Vault

User Profiles

First launch:

Create Profile

Required:

- Username
- Password

Authentication

Supported:

- Local Authentication
- Multiple Profiles

Vault Encryption

Encrypt:

- API Keys
- SSH Passwords
- SSH Keys
- Access Tokens

Algorithms:

- Argon2
- AES-256

Profile Export

Export:

profile.aisprofile

Encrypted archive.

Profile Import

Import profile into another device.

Restore:

- API Keys
- AI Settings
- SSH Hosts
- Preferences

Core Modules

Scanner Engine

Responsibilities:

- Scan disks
- Scan folders
- Calculate sizes
- Collect metadata

Support:

- Windows
- Linux

Remote Scanner

Linux

SSH Support

Scan:

- /
- /var
- /tmp
- /var/log
- Docker Volumes

Windows

Support:

- WinRM
- Network Shares

Scan:

- Temp
- Logs
- User Profiles
- Backup Directories

AI Analysis Engine

Analyze:

- Large files
- Old files
- Temporary files
- Logs
- Backups
- Archives
- ISO files
- Cache files

Generate:

- Recommendations
- Risk Scores
- Cleanup Plans

AI Explain Mode

User can ask:

Why is this folder marked for cleanup?

AI must explain:

- Reason
- File types
- Last usage
- Potential savings
- Risk level

Rules Engine

Built-in logic without AI.

Examples:

- Temp older than 30 days
- ISO older than 180 days
- Log larger than 5 GB
- Backup older than 365 days

AI enriches results.

Duplicate Finder

Methods:

- SHA256
- MD5

Detect:

- Exact duplicates
- Similar files

Cleanup Advisor

Features:

- Preview
- Dry Run
- Safe Delete
- Recycle Bin

Never delete automatically.

Forecast Engine

Predict:

- Disk growth
- Storage trends
- Future disk exhaustion

Display:

- Days remaining
- Growth charts

SRE Storage Analytics

Linux Analytics

Analyze:

- /var/log
- /tmp
- /var/tmp

- Docker
- Podman
- Kubernetes Volumes

Detect:

- Log growth
- Missing rotation
- Oversized containers
- Storage bottlenecks

Generate:

Storage Health Score

Windows Analytics

Analyze:

- Windows Temp
- IIS Logs
- Windows Logs
- Crash Dumps
- Backup Folders

Detect:

- Large logs
- Stale dumps
- Obsolete backups

Generate:

Storage Health Score

User Interface

Login Screen

- Login
- Register
- Import Profile

Dashboard

Display:

- Disk Usage
- Storage Health Score
- Potential Savings
- AI Recommendations

Analysis

Display:

- Scan Results
- Folder Breakdown
- AI Insights

Cleanup

Display:

- Cleanup Candidates
- Risk Levels
- Savings Estimate

Duplicates

Display:

- Duplicate Groups
- Potential Savings

Forecast

Display:

- Growth Trends
- Remaining Capacity

SRE Analytics

Display:

- Storage Health
- Log Analysis
- Cleanup Opportunities
- Infrastructure Recommendations

Settings

Configure:

- AI Providers
- Security
- SSH
- Scan Rules
- Appearance

Database

Tables:

users

profiles

ai_providers

ssh_hosts

scan_history

analysis_results

cleanup_history

duplicate_results

forecast_history

settings

Project Structure

AIStorageAdvisor/

app/

core/

modules/

providers/

security/

database/

ui/

assets/

profiles/

reports/

logs/

exports/

Portable Builds

Windows

Format:

AIStorageAdvisor.exe

Requirements:

- Portable
- No installation required

Build:

PyInstaller

Linux

Formats:

- AppImage
- Portable Binary

Requirements:

- No installation required

Development Roadmap

MVP 0.1

- User Profiles
- Security Vault
- Disk Scanner
- Dashboard
- Ollama Support
- LM Studio Support
- AI Explain Mode

Version 0.5

- Rules Engine
- Cleanup Advisor
- Duplicate Finder

Version 1.0

- Forecast Engine
- Cloud AI Providers
- Profile Export/Import

Version 1.5

- SSH Linux Scanner
- Windows Remote Scanner
- Network Shares

Version 2.0

- Advanced SRE Storage Analytics
- Storage Health Scoring
- Enterprise Reports

Non-Functional Requirements

- Portable
- Offline-first
- Privacy-first
- No telemetry
- Local AI support
- Fast scanning
- Secure credential storage
- Cross-platform

Vision

AI Storage Advisor should become an intelligent storage analysis platform for both home users and SRE/DevOps engineers.

The application must provide actionable insights instead of raw disk statistics and should work equally well with local AI models and cloud AI providers.